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Early vs. late rate of torque development: Relation with maximal strength and influencing factors

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ABSTRACT

We re-examined the relationship between rate of torque development (RTD) and maximal voluntary contractions (MVC) torque, and investigated some possible neuromuscular determinants of early ($\leq$100 ms) and late ($\geq$200 ms) RTD. Seventeen healthy men performed maximal explosive isometric knee extensions at five joint angles, from which MVC torque, RTD at different time intervals (50–250 ms), and early quadriceps EMG activity (EMG_{50}) were evaluated. Quadriceps muscle thickness (MT) was quantified by longitudinal ultrasonography. The relationship between MVC torque, EMG_{50} and MT against RTD was assessed with Pearson's and repeated measures correlation coefficients. Moderate-to-strong correlation coefficients were observed between MVC torque and RTD (r = 0.50–0.88, $p < 0.001$), with stronger relationships for late RTD than for early RTD. Weak-to-strong correlation coefficients were observed amongst RTD and EMG_{50} (r = 0.37–0.83, $p < 0.001$), with stronger relationships for early RTD than for late RTD. Only late RTD was significantly correlated with MT, though only moderately (r = 0.50–0.52, $p < 0.05$). These findings suggest that early and late knee extension RTD are potentially governed by different neuromuscular factors. Neuromuscular activation seems to have a greater influence on early RTD than on late RTD, and vice versa for muscle mass.

1. Introduction

The maximal force-generating capacity of the knee extensor muscles is of critical importance for daily living and sport activities. This capacity is usually evaluated by dynamometry as the highest isometric torque of a slow ramp and hold maximal voluntary contraction (MVC) conducted at a predefined knee angle (Wilson and Murphy, 1996; McMaster et al., 2014). The main determinant of MVC torque is the total amount of muscle mass, as evidenced by strong and consistent correlations with imaging-derived variables such as muscle cross-sectional area (Franchi et al., 2018; Maden-Wilkinson et al., 2020), volume (Franchi et al., 2018; Miyatani et al., 2002) and thickness (Ando et al., 2015; Franchi et al., 2018). Besides maximal strength, the ability to generate force within a short time interval (i.e. explosive strength) is increasingly considered by researchers and practitioners due to its functional relevance (Buckthorpe and Roi, 2017; Maffiuletti et al., 2016). Explosive strength is often characterized as the rate of torque development (RTD) at different time intervals from the onset of an explosive MVC (Maffiuletti et al., 2016). Due to the recent success of RTD, different studies have been conducted to elucidate the main methodological and physiological aspects of this promising variable (Maffiuletti et al., 2016; Rodríguez-Rosell et al., 2018; Thompson et al., 2018; Tillin et al., 2013), particularly in relation with time-interval specificities.

RTD is often classified as "early" and "late" for time intervals $\leq$100 ms and $\geq$200 ms, respectively (Andersen and Aagaard, 2006; Folland et al., 2014). Early RTD seems to be primarily influenced by mechanisms in relation with neural activation transmitted by motor neurons to muscles (Del Vecchio et al., 2019b; Folland et al., 2014; Maffiuletti et al., 2016). On the other hand, because late RTD is strongly related to maximal strength (Andersen and Aagaard, 2006), it is influenced by similar underlying mechanisms (Blazevich, 2006; Maden-Wilkinson et al., 2020). As a matter of fact, an ultrasound-derived measure of

Abbreviations: EMG, electromyographic; EMG_{50}, early electromyographic activity; MT, muscle thickness; MVC, maximal voluntary contraction; RF, Rectus femoris; RMS, root mean square; RTD, rate of torque development; VI, Vastus intermedius; VL, Vastus lateralis; VM, Vastus medialis.

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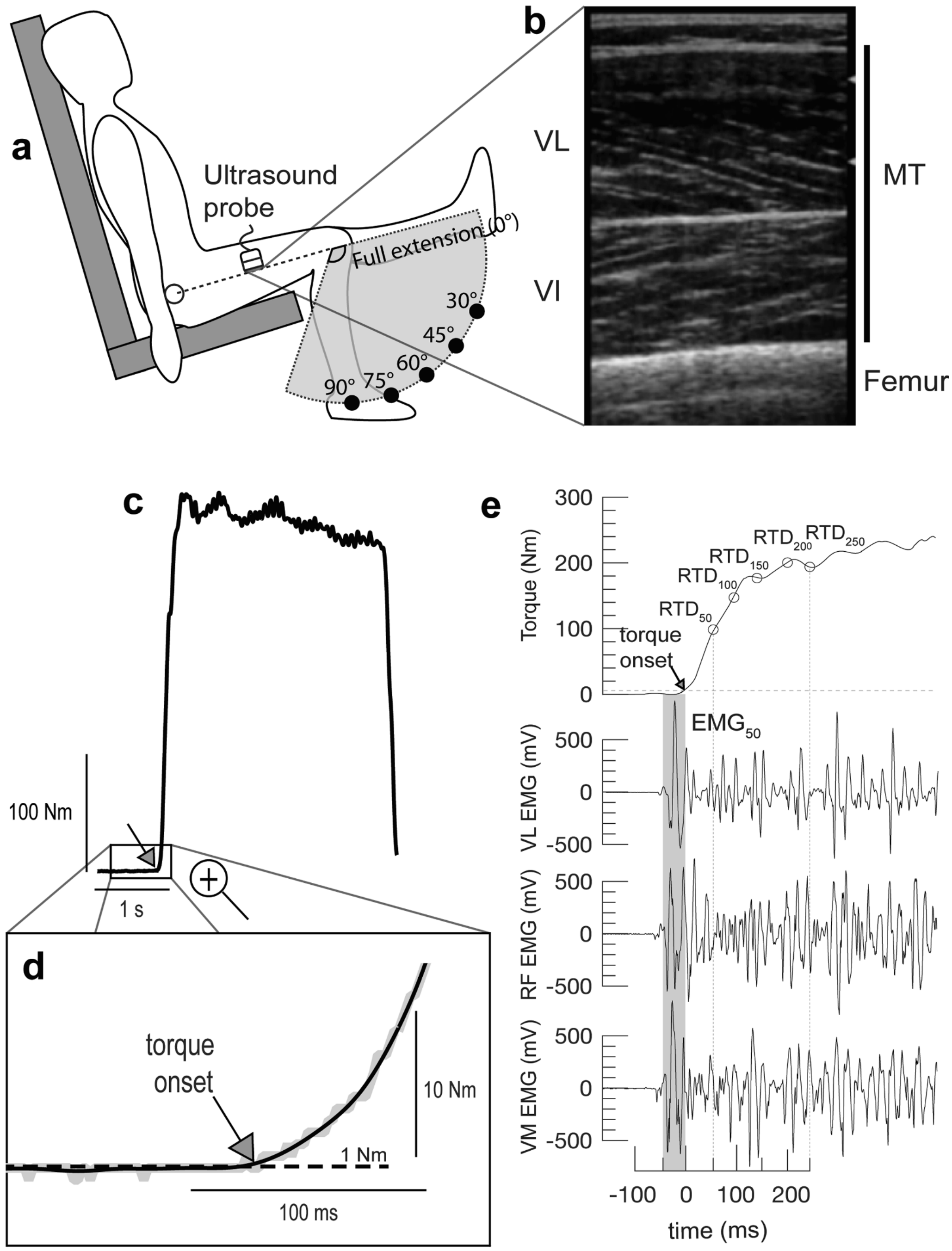

Fig. 1. Overview of the experimental set-up. a) Representative set-up showing the position of the participant on the dynamometer, the five knee angles at which the measurements were performed and the position of the ultrasound probe on the lateral aspect of the thigh. b) Representative longitudinal ultrasound image, from which quadriceps muscle thickness was quantified. c) Representative torque signal recorded during the maximal voluntary explosive isometric contractions, from which MVC torque and RTD variables (see also e) at different time intervals from torque onset (see also d) were quantified. e) Torque signal (both raw and filtered) at the very beginning of the contraction is magnified to show torque onset, which corresponded to a threshold of 1 Nm above baseline. e) Representative torque and EMG signals at the onset of the contraction, from which RTD variables and early EMG activity (EMG_{50} – shaded area) were quantified.

quadriceps muscle mass - muscle thickness (MT) - has recently been found to correlate positively with late RTD (Coratella et al., 2020; Wagle et al., 2017; Zaras et al., 2016).

Thus, early and late RTD appear to be two somewhat different measures of explosive strength. Considering the already-reported similarities between late RTD and MVC torque, it could even be questioned whether late RTD actually represents a valid indicator of explosive strength. Therefore, the two main aims of this methodological study were (1) to re-examine the relationship between RTD obtained at different time intervals and MVC torque and (2) to investigate some possible neuromuscular determinants of early and late RTD. Accordingly, we hypothesized that (1) late RTD would be better correlated to MVC torque than early RTD, and (2) early and late RTD would be correlated respectively to early EMG activity and MT. One unique aspect of this study is that EMG activity was quantified during a short time interval (50 ms) at contraction onset, when the amount of amplitude cancellation is presumably lower compared to later phases (due to a lower number of active motor units) (Keenan et al., 2005). Also, knee extensor MVC torque and RTD, as well as early quadriceps EMG activity, were measured at five different knee joint angles and repeated measures correlation analyses were used to account for the well-known dependence of joint torque on muscle-tendon unit length (Herzog, 2017; MacIntosh, 2017).

2. Methods

2.1. Participants

Seventeen healthy, physically active men (mean age SD: 28 ± 6 years old, body mass: 81 ± 10 kg, height: 177 ± 5 cm, physical activity level: 5345 ± 3336 min/week [International Physical Activity Questionnaire]) volunteered to participate in the study. The exclusion criteria

Table 1
MVC torque, RTD and EMG_{50} values across knee flexion angles.

	30° (1)	45° (2)	60° (3)	75° (4)	90° (5)	p value	η_p^2
MVC torque (Nm)	169 ± 31 (2,3,4,5)	216 ± 47 (1,3,4)	258 ± 59 (1,2,5)	270 ± 54 (1,2,5)	225 ± 40 (1,3,4)	<0.001	0.73
RTD_{50} (Nm/s)	712 ± 676 (3,4)	965 ± 787 (NS)	1321 ± 796 (1)	1308 ± 656 (1)	909 ± 565 (NS)	<0.001	0.23
RTD_{100} (Nm/s)	924 ± 323 (3,4)	1200 ± 454 (NS)	1341 ± 456 (1,5)	1306 ± 380 (1,5)	1045 ± 395 (3,4)	<0.001	0.30
RTD_{150} (Nm/s)	832 ± 220 (2,3,4,5)	1088 ± 287 (1)	1143 ± 238 (1,5)	1123 ± 264 (1)	964 ± 256 (1,3,4)	<0.001	0.39
RTD_{200} (Nm/s)	674 ± 164 (2,3,4,5)	883 ± 226 (1)	988 ± 201 (1,5)	971 ± 224 (1,5)	838 ± 187 (1,3,4)	<0.001	0.57
RTD_{250} (Nm/s)	577 ± 139 (2,3,4,5)	754 ± 176 (1,3)	848 ± 181 (1,2,5)	851 ± 189 (1,5)	755 ± 147 (1,3,4)	<0.001	0.65
EMG_{50} (%)	0.57 ± 0.41 (3,4)	0.67 ± 0.39 (1,3)	1.00 ± 0.61 (1,2,5)	0.82 ± 0.41 (NS)	0.68 ± 0.40 (3)	<0.001	0.21

Values are mean ± SD. The p values and η_p^2 refer to the knee angle effect. Superscripts represent the pairwise statistical difference (all $p < 0.05$): (1) different from 30°, (2) different from 45°, (3) different from 60°, (4) different from 75°, (5) different from 90°, (NS) Non-significantly pairwise differences.

were: history of lower limb and/or spine orthopaedic injuries, any kind of pain, and cardiovascular disease. The study was approved by the local ethics committee for human experiments and all participants provided written informed consent prior to participating.

2.2. Experimental protocol

All measurements were performed during a single testing session lasting ~60 min. Participants were asked to complete maximal voluntary explosive isometric contractions of the knee extensors at five different knee joint angles, from which MVC torque, RTD at different time intervals, and early EMG activity were recorded. Subsequently, quadriceps MT was measured by longitudinal ultrasonography at one knee angle and in a resting state. All measurements were conducted unilaterally on the dominant side, which was self-defined as the preferred kicking leg (16 right and 1 left).

2.3. Assessment of MVC torque and RTD

Participants were positioned on an isokinetic dynamometer (Humac Norm II, CSMI, Stoughton, MA, USA) with both hips at ~90°, the trunk stabilized by seat belts, and the knee aligned with the dynamometer rotational axis. Maximal voluntary explosive isometric contractions of the knee extensors were conducted at five different knee angles (30, 45, 60, 75, and 90° of flexion) (Fig. 1A), randomly presented. Each contraction lasted ~3 s and three attempts separated by 30 s of rest were conducted at each angle. A 3-min rest period was held between angles to minimize neuromuscular fatigue. Subjects were consistently instructed to produce force "as fast and hard as possible" and to keep the maximum effort whenever the verbal instruction given by the tester persisted. To minimize the dampening effect occurring at the interface between the shin and the lever, and simultaneously to avoid pain, a standard soccer hard shin guard was used (de Ruiter et al., 2004). Prior to the maximal trials, subjects completed a familiarization procedure to learn how to execute the explosive contractions. Additionally, at least two submaximal contractions were performed at each angle as a warm-up. Real-time visual feedback of the torque signal was continuously provided to the participants by means of a screen. The torque signal provided by the dynamometer was sampled at 1 kHz using an external analog-to-digital converter (EMG 830c®, EMG System do Brasil, São José dos Campos, SP, Brazil), and was recorded concomitantly with EMG signal for further processing.

Torque and EMG signals were processed by a custom-built software developed in Matlab (Mathworks Inc., Natick, MA). The torque signal was filtered using a fourth-order, 50 Hz cut-off frequency low-pass Butterworth zero-lag filter. Pilot measurements demonstrated that this cut-off frequency was enough to smooth the signal without shifting the onset of the contraction (Fig. 1C and D). Torque onset was defined by a fixed torque threshold of 1 Nm and was visually determined by the same evaluator with a dashed line created by the analysis software (Fig. 1C and D) (de Ruiter et al., 2007; Thompson, 2019). At each angle, the torque signal was systematically corrected for the passive torque using the average torque produced at the transducer during a baseline resting period of 500 ms. MVC torque was quantified as the single highest torque value recorded throughout the maximal explosive contractions. RTD was calculated as the slope of the torque-time trace from 50 to 250 ms in steps of 50 ms (RTD_{50}, RTD_{100}, RTD_{150}, RTD_{200}, RTD_{250}; Fig. 1E) from torque onset (Maffiuletti et al., 2016). For each knee angle, only the trial with the highest MVC torque was used for all the analyses.

2.4. Assessment of EMG activity

The skin was shaved with a razor and carefully cleaned with alcohol before electrode placement. Pairs of silver chloride, circular (recording diameter: 10 mm) surface electrodes (RedDot®, 3 M Corp, Saint Paul, MN, EUA) were positioned over the rectus femoris (RF), vastus medialis (VM) and vastus lateralis (VL) muscle bellies according to standard recommendations (Hermens et al., 2000), with an inter-electrode distance of 20 mm. The reference electrode was placed on the medial epicondyle of the humerus. The EMG signal was amplified (gain: 1000), band-pass filtered (20–500 Hz), and recorded at a sampling frequency of 1 kHz by means of a standard device for biological signal recording (EMG 830c®, EMG System do Brasil, São José dos Campos, SP, Brazil).

EMG signals were filtered for the reduction of power grid noise using the technique proposed by Mello et al. (2007). Early EMG activity was calculated as the root mean square value of a 50-ms window immediately preceding torque onset (EMG_{50}) (Fig. 1E). We considered this EMG variable for different reasons: (1) to minimize the amount of amplitude cancellation, as the number of active motor units was presumably lower compared to later contraction phases; (2) to account for possible differences in electromechanical delay between muscles, as the EMG signal was analysed in relation with torque rather than EMG onset; (3) to maximize the association between EMG and torque, as EMG_{50} was the EMG variable showing the strongest correlations with RTD variables in pilot testing. The EMG_{50} values were normalized to the maximal root mean square value recorded during a 500-ms interval around MVC torque (250 ms before and 250 ms after). All these calculations were conducted separately for the three investigated muscles, and finally, the average EMG_{50} ([VL + RF + VM]/3) was retained to represent early quadriceps EMG activity.

2.5. Assessment of MT

Ultrasound images were captured using a 7.5 MHz, 40-mm linear array probe in B-mode (SDD800, ALOKA, Japan). Participants were seated on the dynamometer with the knee at 30° of flexion in a completely relaxed state. The images were acquired longitudinally at one half of the distance between the greater trochanter and the superior edge of the patella (Blazevich, 2006; Kumagai et al., 2000). Longitudinal-view quadriceps MT, which includes VL and vastus intermedius (VI) muscles, was determined as the linear distance between the upper aponeurosis of the VL muscle and the upper visible border of the femur (Fig. 1B). Measurements were made using the ImageJ 1.51 software (National Institutes of Health, Bethesda, MD, EUA). Each

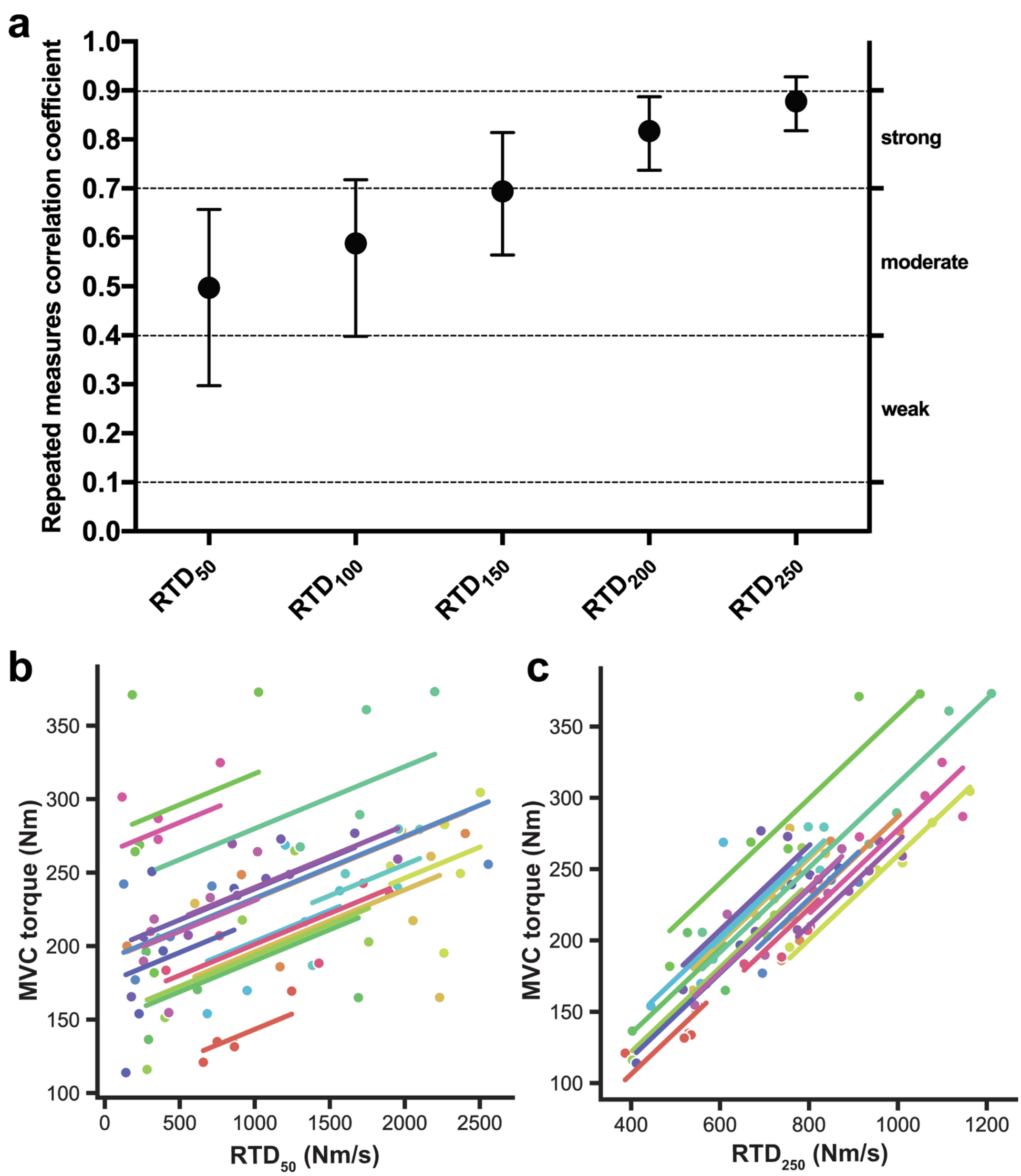

Fig. 2. a) Repeated measures correlation coefficients between MVC torque and RTD variables across knee angles. The benchmarks and 95% confidence intervals are also shown. All correlation coefficients were significant ($p < 0.001$). b) Full scatterplot of repeated measures correlation analysis between MVC torque and RTD_{50} (each subject is represented by a different colour). For each subject, the different points correspond to the five knee angles, and the respective linear adjustments are also shown. c) Full scatterplot of repeated measures correlation analysis between MVC torque and RTD_{250}.

individual image was measured three times in the proximal, central, and distal portions by the same investigator, and the average MT was retained to represent quadriceps MT.

2.6. Statistical analyses

The effect of knee angle on MVC torque, RTD variables, and EMG_{50} was investigated using a one-way ANOVA for repeated measures. If sphericity could not be assumed, the Greenhouse-Geisser correction for the degree of freedom was employed. The Bonferroni test for pairwise comparison was used when necessary. The strength of the linear relationship between MVC torque or EMG_{50} and RTD obtained at different time intervals was assessed with repeated measures correlation coefficients across knee angles (Bakdash and Marusich, 2017). The strength of the linear relationship between MT and 30°-knee flexion RTD obtained at different time intervals (as well as MVC torque) was assessed with Pearson's correlation coefficients. Both repeated measures and Pearson's correlation coefficients were calculated with the Python's package "pingouin" (Vallat, 2018) using the function "rm_corr" (Bakdash and Marusich, 2017) and "corr", respectively. Correlation coefficients were arbitrarily considered as negligible (0.00–0.09), weak (0.10–0.39), moderate (0.40–0.69), strong (0.70–0.89), and very strong (0.90–1.00) (Schober et al., 2018). The effect size was assessed by η_p^2 and classified as small (0.01), medium (0.06), and large (0.14) (Cohen, 1988). The level of significance was set to $p < 0.05$ for all the procedures.

3. Results

The mean ± SD MVC torque, RTD at different time intervals, and EMG_{50} values obtained at the different knee angles are displayed in Table 1. A significant knee angle effect was observed for all these

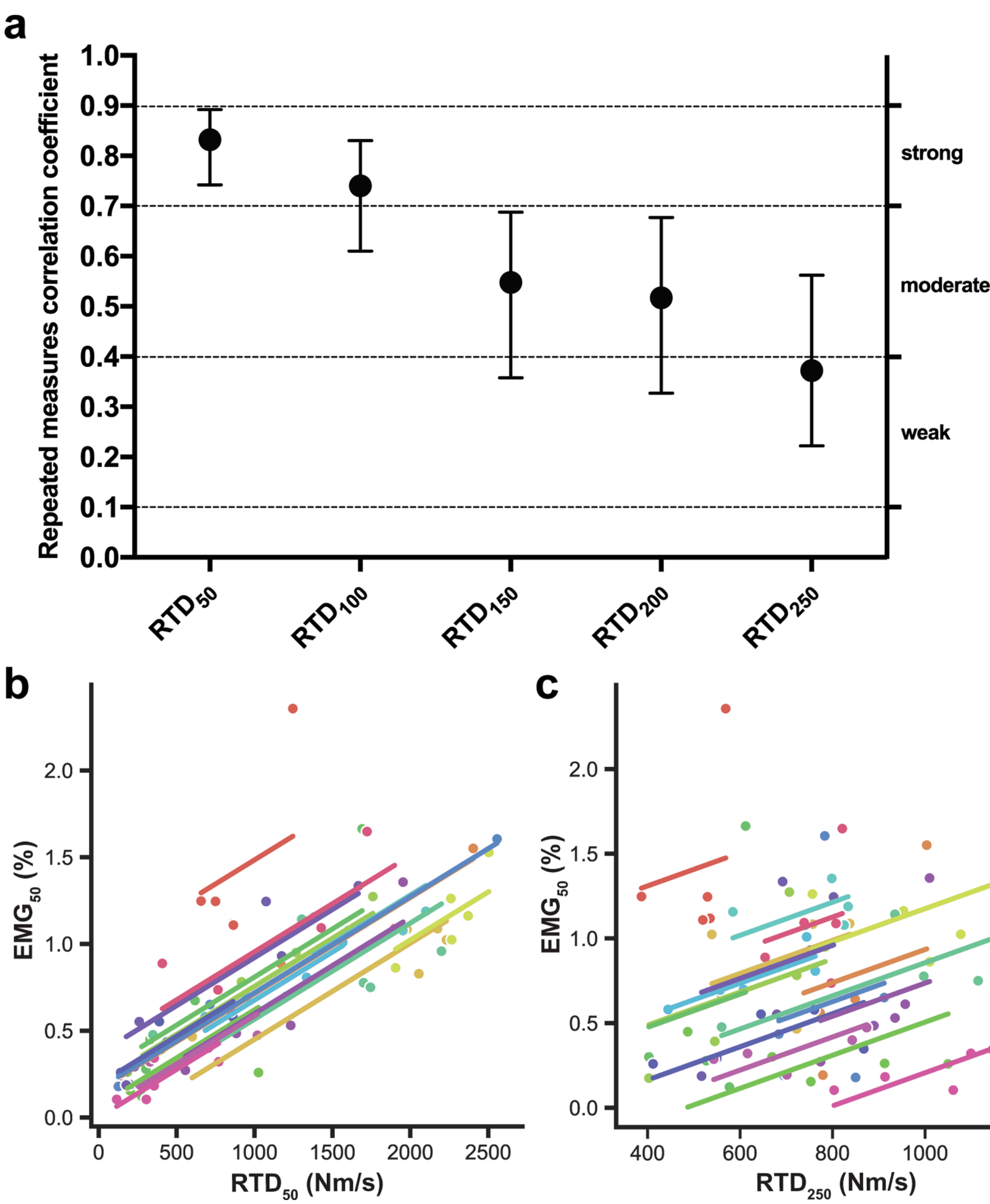

Fig. 3. a) Repeated measures correlation coefficients between EMG_{50} and RTD variables across knee angles. The benchmarks and 95% confidence intervals are also shown. All correlation coefficients were significant ($p < 0.001$). b) Full scatterplot of repeated measures correlation analysis between EMG_{50} and RTD_{50} (each subject is represented by a different colour). For each subject, the different points correspond to the five knee angles, and the respective linear adjustments are also shown. c) Full scatterplot of repeated measures correlation analysis between EMG_{50} and RTD_{250}.

variables ($p < 0.001$). The pairwise comparisons suggest similarities in the differences between knee flexion angles for the ensemble of the variables (i.e., the highest and lowest values were observed at 60° and 30° of flexion, respectively), and more particularly so for MVC torque and late RTD.

Moderate-to-strong correlation coefficients were observed between MVC torque and RTD variables across knee angles ($p < 0.001$; Fig. 2A). The strength of the linear relationships increased with increasing RTD time intervals: correlation coefficients were moderate for RTD_{50} (Fig. 2B), RTD_{100} and RTD_{150}, and strong for RTD_{200} and RTD_{250} (Fig. 2C).

Weak-to-strong correlation coefficients were observed between EMG_{50} and RTD variables across knee angles ($p < 0.001$; Fig. 3A). The strength of the linear relationships decreased with increasing RTD time intervals: correlation coefficients were strong for RTD_{50} (Fig. 3B) and RTD_{100}, moderate for RTD_{150} and RTD_{200}, and weak for RTD_{250} (Fig. 3C).

The mean ± SD quadriceps MT was 51.0 ± 6.2 mm. MT was significantly correlated to RTD_{200} ($p = 0.04$) and RTD_{250} ($p = 0.03$), though only moderately (Fig. 4AC), but not to lower time-interval RTD variables (Fig. 4AB). The strength of the linear relationships increased with increasing RTD time intervals. MT was also significantly correlated to MVC torque, with a strong correlation coefficient ($r = 0.72$, $p < 0.001$, 95% confidence interval = 0.37–0.89).

4. Discussion

The main findings of this study were that late RTD was better correlated with MVC torque and quadriceps muscle mass than early RTD, with correlation coefficients increasing proportionally with increasing time interval. On the other hand, early RTD was better correlated with early quadriceps EMG activity than late RTD, with correlation coefficients decreasing proportionally with increasing time interval. These results, taken together with those of other studies

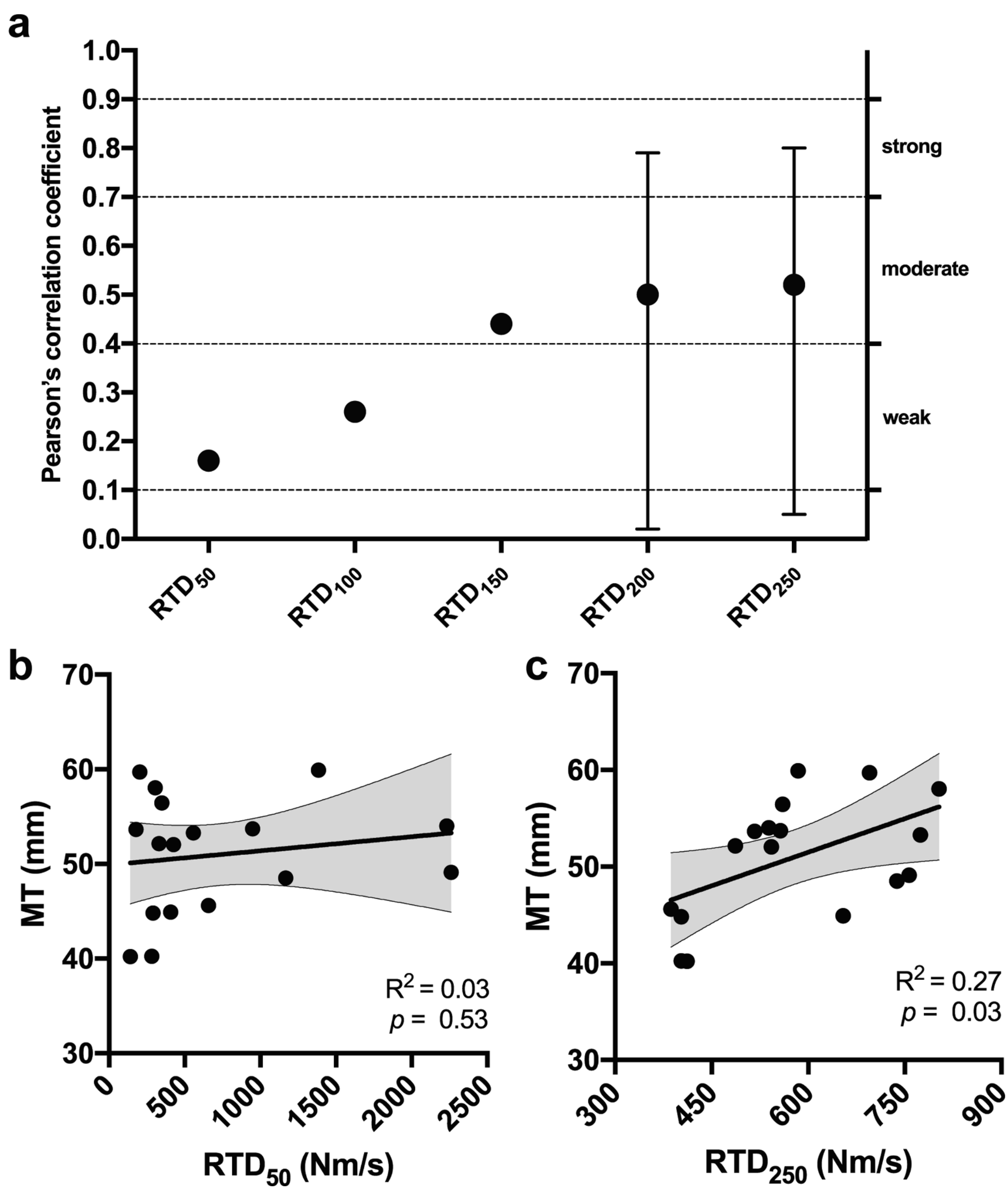

Fig. 4. a) Pearson's correlation coefficients between quadriceps MT and RTD variables obtained at 30° of knee flexion. The benchmarks and 95% confidence intervals are also shown. Open and closed circles indicate respectively non-significant and significant correlation coefficients ($p < 0.05$). b) Scatterplot of Pearson's correlation analysis between MT and RTD_{50}. The linear adjustment and 95% confidence intervals are also shown. c) Scatterplot of Pearson's correlation analysis between MT and RTD_{250}.

(Andersen and Aagaard, 2006; Del Vecchio et al., 2019b; Folland et al., 2014; Maffiuletti et al., 2016), strongly support the notion that early and late RTD are influenced by different neuromuscular factors.

We observed moderate-to-strong correlation coefficients between RTD and MVC torque. This is in full agreement with the results of Andersen and Aagaard (2006), who observed proportional increases in correlation coefficients with increasing RTD time intervals from 50 to 250 ms. Even though the strong association between maximal strength and RTD is not a novelty (Andersen and Aagaard, 2006; Mirkov et al., 2004), data were collected at five different knee angles and analysed with repeated measures correlations in our current study vs. a single joint angle and simple pairwise correlations in the previous studies. We chose this method to account for the knee angle-torque relationship, and consequently for the intrinsic changes of muscle-tendon unit mechanical properties and net torque production capacity as a function of muscle length (Herzog, 2017; MacIntosh, 2017). Interestingly, not just late RTD, but also early RTD was significantly correlated with MVC torque, suggesting that the link between maximal and explosive strength is probably stronger than generally believed. On the other hand, the strong relationship observed between late RTD and MVC torque further confirms the redundancy of these variables.

The relationship between MT and late RTD (i.e., RTD_{200} and RTD_{250}) was significant, alike MVC torque, but contrary to earlier phase RTD variables. These results suggest that - similarly to maximal strength (Maden-Wilkinson et al., 2020) - the amount of quadriceps muscle mass is one of the main determinants of late knee extension RTD, likely in relation with the number of sarcomeres in parallel (Blazevich, 2006; Lieber and Fridén, 2000). Late RTD has already been found to be positively correlated to both VL (Zaras et al., 2016) and VI (Coratella et al., 2020) MT. In particular, Coratella et al. (2020) observed significant correlations for RTD intervals comprised between 100 and 250 ms ($r = 0.55$–0.69) but not for shorter time epochs ($r = 0.27$–0.46). However, none of the other quadriceps muscle heads (including the VL) displayed significant correlations in this previous study. Even if we quantified MT of both VL and VI - which together represent ~63% of the total quadriceps volume (Maden-Wilkinson et al., 2020; Miyatani et al., 2002) - and ultrasound-derived MT is strongly related to magnetic resonance imaging-derived measures of quadriceps muscle mass (at least for the VL) (Franchi et al., 2018), our current results remain to be confirmed using more valid indicators such as muscle physiological cross-sectional area or volume. Also, it should be kept in mind that other muscular factors not addressed here influence late (but also early) RTD such as architectural properties (Blazevich et al., 2009; Zaras et al., 2016), muscle fiber type distribution (Harridge et al., 1996; Methenitis et al., 2019), and muscle-tendon stiffness (Ando and Suzuki, 2019; Bojsen-Møller et al., 2005).

We observed strong correlations between early EMG activity and early RTD that we interpret in relation with the specificity of the neural activation transmitted by motor neurons to muscles during explosive contractions. The first evidence of an association between motor unit behaviour and RTD was provided by Desmedt and Godaux (1977) who observed extremely high instantaneous motor unit discharge rates (and decreased recruitment threshold) at the onset of explosive "ballistic" contractions. More recently, a high-density surface EMG decomposition study has confirmed the causal association between RTD and initial motor unit discharge rate and recruitment speed, particularly for the first 35–40 ms of activity (Del Vecchio et al., 2019b). Thus, motor unit behaviour - discharge rate in particular - at the onset of explosive contractions can confidently be considered as a major determinant of early RTD (Del Vecchio et al., 2019b, 2019a; Dideriksen et al., 2020; Duchateau and Baudry, 2014). We evaluated EMG amplitude with classical bipolar electrodes in this study which, despite well-known shortcomings (Farina et al., 2010; Vigotsky et al., 2017), represents the sum of muscle fibre action potentials (muscle activation) and thus is influenced by motor unit recruitment and discharge rate. Interestingly, the strength of the relationship between early EMG activity and RTD decreased with increasing RTD time intervals in our study, with a time-course comparable to the one observed by Del Vecchio et al. (2019b) for an estimate of the neural drive to the muscle (i.e., average number of discharges per motor unit per second).

The present results have important implications for researchers and practitioners seeking to optimize the assessment of RTD and to improve explosive strength. For example, the time interval(s) for RTD calculation should be better justified and early RTD ($\leq$100 ms) should be consistently distinguished from late RTD ($\geq$200 ms). Early RTD may preferentially be used for acute or chronic interventions aimed at modifying neuromuscular activation - despite limited reliability (Buckthorpe et al., 2012) - while late RTD may be more appropriate when maximal strength and/or muscle mass changes are expected. Speculatively, RTD_{150} can be considered as an intermediary measure of explosive strength. Finally, and more importantly, future studies should probably refrain from considering late RTD as an indicator of explosive strength and from presenting both MVC torque and late RTD due to their redundancy.

The current study has some limitations that need to be acknowledged. Most of our findings are based on multivariate and simple correlational analyses that, despite having been strengthened by the inclusion of torque and EMG data gathered at different knee angles, should be interpreted with caution. Also, we only considered few, selected and global neuromuscular variables (EMG_{50} and MT) - due to their accessibility - while the explosive force-generating capacity is actually influenced by numerous other neural and muscular factors (Bojsen-Møller et al., 2005; Dideriksen et al., 2020; Duchateau and Baudry, 2014; Duchateau and Enoka, 2011; Folland et al., 2014; Lanza et al., 2019; Zaras et al., 2016). Finally, quadriceps MT was quantified exclusively at one knee joint angle and in passive conditions. However, since both VL and VI muscles are pennate and thus can be modelled as preconized by the parallelogram model (Narici et al., 2016), their MT would not change as a function of muscle-tendon unit length. This assumption was confirmed by a pilot experiment conducted with 30 subjects, in which quadriceps MT (VL + VI) did not differ significantly across the 30–90° knee flexion angle range.

In conclusion, we provide some experimental evidence that early and late knee extension RTD are potentially governed by different neuromuscular factors. Neuromuscular activation seems to contribute more to torque generation in the early phase of an explosive contraction - as witnessed by EMG results - while the contribution of muscle mass may be more influential in later phases (including maximal strength) - as witnessed by MT results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors' contributions

All authors contributed equally to the conception of the study, data analysis, data interpretation and manuscript preparation. Both authors read and approved the final version of the manuscript.

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